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Enabling single-molecule-derived blue room-temperature phosphorescence in polymers via melt processing

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Highlights

- Melt processing yields single-molecule blue RTP (455 nm) in hydrogen-bonding polymers.
- Tunable blue–green RTP and near-white emission (CIE 0.33, 0.36) via doping control.
- Thermoplastic RTP materials enable 3D printing, molding, and anti-counterfeiting applications.

Abstract

Outline

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Pure organic blue room-temperature phosphorescence (RTP) remains challenging due to aggregation-induced quenching. We introduce a scalable melt-processing strategy for fabricating polymer-based blue RTP materials by dispersing phosphors (e.g., 4,4'-dihydroxydiphenyl sulfone, SDP) into hydrogen-bonding polymer matrices, i.e., ethylene-vinyl alcohol copolymer (EVOH). The high temperature and shear forces during processing enable uniform molecular dispersion, while hydrogen bonding stabilizes triplet excitons, yielding blue RTP at 455 nm from isolated molecules, which is distinct from the green emission of aggregated phosphors in solution-processed films. The approach is highly tunable: varying doping concentration shifts emission from blue to green, and co-doping complementary phosphors produces white RTP with CIE coordinates (0.33, 0.36). The materials also support injection molding, 3D printing, and UV-responsive anti-counterfeiting. This work addresses the scarcity of stable blue RTP materials and offers a practical, scalable route for controlled luminescence, expanding applications in optoelectronics and encryption.

Graphical abstract

A scalable melt-processing strategy addresses aggregation-induced quenching in pure organic blue room-temperature phosphorescence (RTP). By extruding phosphors (e.g., SDP) with hydrogen-bonding polymers (e.g., EVOH) via a single-screw extruder, high temperature and shear forces enable uniform molecular dispersion, while matrix-phosphor hydrogen bonding stabilizes triplet excitons—yielding single-molecule-derived blue RTP at 455 nm (distinct from aggregated green emission in solution processing). This approach is highly tunable: adjusting doping concentration induces blue-to-green transitions, and co-doping complementary phosphors (p-ASA/NA) produces near-standard white RTP (CIE: 0.33, 0.36). The materials support injection molding and 3D printing, enabling UV-responsive anti-counterfeiting via time-dependent emission transitions. This work provides a facile, scalable route to stable blue RTP materials for

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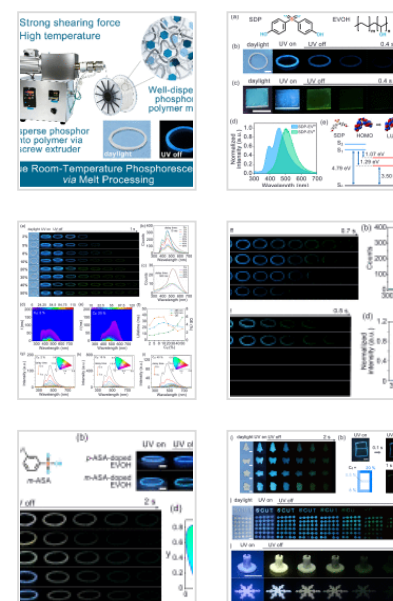
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Figures (6)



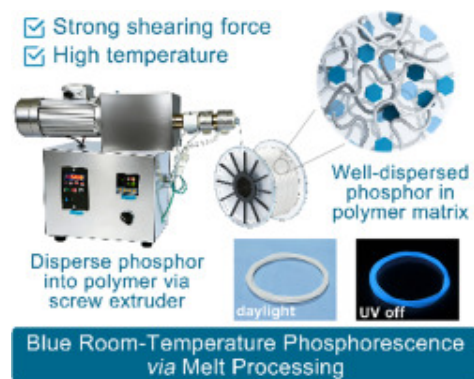
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1. Introduction

Metal-free room-temperature phosphorescence (RTP) materials have garnered growing attention over the past decades, driven by their promising applications in displays, sensors, biomedicine, and optical communication.[1], [2], [3], [4] While significant progress has been made in developing RTP materials with tunable emission colors, the fabrication of pure organic blue RTP materials remains a challenge.[5], [6], [7], [8] Blue phosphorescent emitters typically suffer from poor stability and relatively long triplet exciton lifetimes, which lead to severe performance degradation.[9], [10], [11] Additionally, their wide band gaps and high triplet exciton energies make it difficult to identify suitable host materials and carrier transport materials for designing efficient, stable blue devices.[12], [13] A further complication is that the aggregation of phosphors often causes red-shifted emission or triplet exciton annihilation quenching, further hindering the development of high-performance blue RTP systems.[4], [14]

Polymer-based RTP materials offer distinct advantages over small-molecule counterparts, including good flexibility, ease of processing, low cost, and environmental resistance.[15], [16], [17], [18] As such, they have emerged as attractive alternatives to low-molecular-weight organic RTP compounds.[19], [20] The most common approach to preparing polymer-based RTP materials involves doping organic phosphors into polymer matrices, which is a strategy that has been widely explored since the first reports of dye-doped phosphorescent polymers in the 1960s.[21], [22], [23] However, most previous studies relied on solution processing to mix phosphors and polymers, which limits the practical utility of these materials. Melt processing, by contrast, is a dominant and scalable technique in the polymer industry, capable of fabricating materials into diverse shapes beyond thin films. Yet, its potential for preparing polymer-based blue RTP materials has been largely underexplored.

Recent advances in organic RTP materials have focused on enhancing intersystem crossing (ISC) and reducing non-radiative transitions through strategies such as crystallization inducement,[24] H-aggregation,[25] co-crystallization,[26] polymerization,[27] and host-guest doping.[17] Despite these efforts, inherent limitations persist, for instance, crystal engineering may dissipate triplet excitons via triplet-triplet annihilation, causing phosphorescence darkening.[28] Notably, blue-shifted phosphorescence emission is achievable in the single-molecule state of phosphors but not in their aggregated state. However, realizing single-molecule blue RTP at room temperature is extremely challenging. For example, dilute solutions of phosphors typically require low temperatures (e.g., 77K) to slow molecular motion and suppress aggregation-caused quenching, which is incompatible with practical applications. While spatial confinement (e.g., via cucurbituril encapsulation, ionic crystal confinement, or tetrahedral structure restriction) has enabled single-molecule blue RTP,[29], [30] the poor processability of organic crystals limits their use in optoelectronics.

To address these gaps, we hypothesized that dispersing phosphors in a rigid polymer matrix at the molecular level with minimal non-radiative transitions could enable blue RTP emissions. For the materials prepared by solution mixing, aggregation of phosphors

takes place during drying process. In contrast, the strong shearing forces and high-temperature fluidity of melt processing offers a means to uniformly disperse phosphors in polymers. This work demonstrates a scalable melt-processing strategy for fabricating polymer-based blue RTP materials with tunable emission, excellent processability, and potential applications in anti-counterfeiting and optoelectronics.

2. Experimental section

2.1. Materials

EVOH (ethylene-vinyl alcohol copolymer, 48 mol% ethylene content) was purchased from Kuraray Co., Ltd. *N,N*-Dimethylformamide (DMF), 4,4'-sulfonyldiphenol (SDP), *p*-aminobenzene sulfonic acid (*p*-ASA), *m*-aminobenzene sulfonic acid (*m*-ASA), and naphthalic anhydride (NA) were supplied by Innochem.

2.2. Measurements and methods

UV-vis absorption spectra were recorded using a Shimadzu UV3600PLUS UV-Vis-NIR spectrophotometer. Fluorescence spectra were obtained with a Shimadzu RF6000 spectrofluorometer. Phosphorescence spectra and steady-state time-resolved emission spectra (TRES) were measured via a fiber optic spectrometer (Maya 2000Pro, Ocean Optics) excited by a standard 365 nm UV lamp. Phosphorescence lifetime decay curves were acquired on a Hitachi F-4700 spectrophotometer with a 365 nm xenon (Xe) light source. In-situ photoluminescence (PL) emission measurements were conducted using a home-made hotstage mounted on an OmniFluo900 spectrophotometer. Fourier-transform infrared (FTIR) spectroscopy was performed with a JASCO FT/IR-6000 spectrometer.

2.3. Preparation of thin films

EVOH was dissolved in DMF at 90 °C to a concentration of 30 mg/mL. SDP was then added at doping ratios (mass/mass) of 2%, 5%, 10%, and 20%, and the mixtures were sonicated for

30min to ensure homogeneous dissolution. The resulting solutions were drop-coated onto clean substrates, and the films were pre-dried on a hotplate at 80 °C to accelerate solvent evaporation, followed by vacuum drying at 80 °C for 10h to remove residual solvent.

2.4. Preparation of filaments

EVOH (10.0g) was dried in a vacuum oven at 60 °C for 10h to eliminate moisture. SDP was physically blended with the dried EVOH at doping ratios (mass/mass) of 2%, 5%, 8%, 10%, 20%, 30%, 40%, and 50%. The blends were extruded into filaments using a single-screw extruder at given processing temperatures.

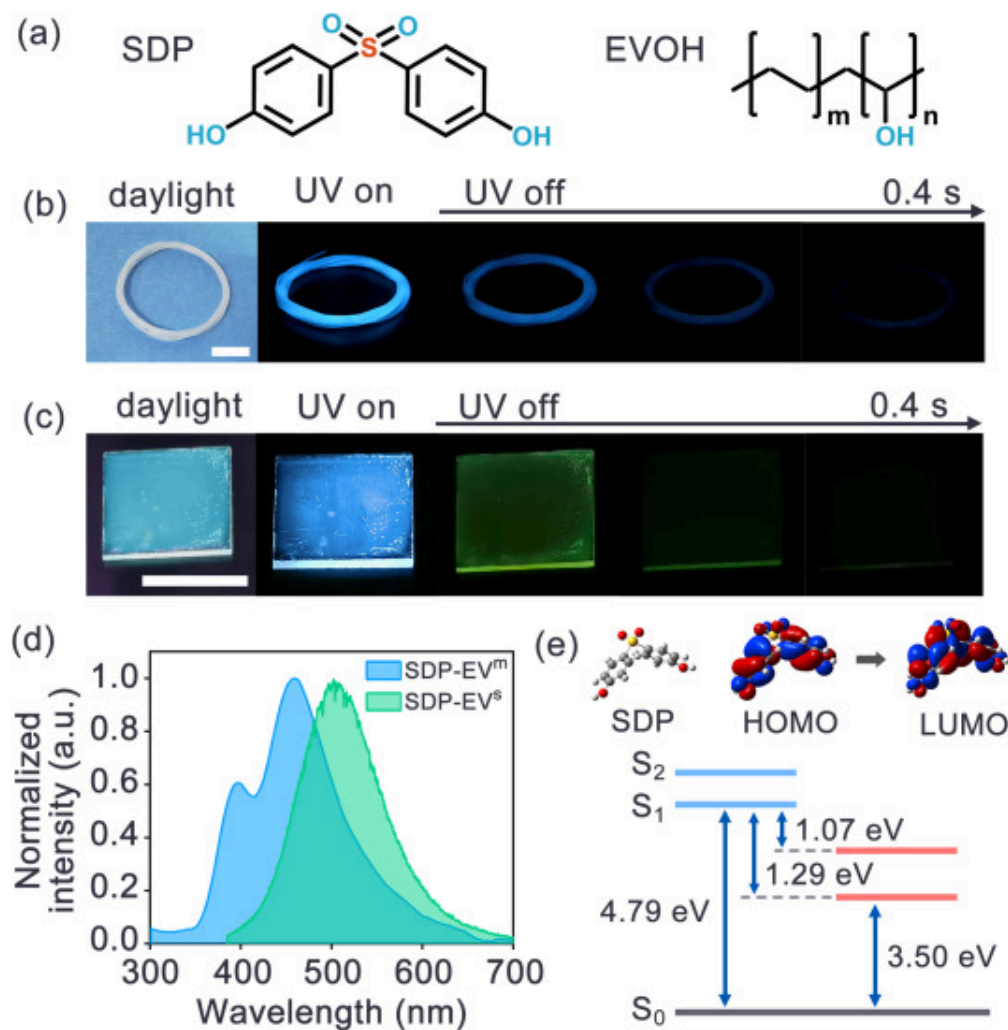
2.5. Density functional theory (DFT) calculations

All DFT calculations were performed using the B3LYP hybrid functional with the 6–311++G(d,p) basis set. Full geometry optimization and vibrational frequency analysis were conducted for seven bisphenol analogues to obtain their ground-state stable structures. Based on the optimized configurations, we calculated the thermodynamic parameters, electronic structure properties (including frontier molecular orbitals and reactivity descriptors), and molecular electrostatic potential maps of the target compounds. The FT-IR and Raman spectra were calculated with a wavenumber scaling factor of 0.9679, and the UV–Vis absorption properties were investigated via time-dependent DFT (TD-DFT) at the same theoretical level.

3. Results and discussion

Herein, we selected SDP as the phosphor and EVOH as the polymeric matrix (Fig. 1a). The sulfone group (O=S=O) in SDP can enable a strong SOC effect to promote the ISC.^[31] The dilute solution of SDP in THF exhibited blue phosphorescent emission at 77K, whereas the SDP powder presented green afterglow at ambient condition after the irradiation (Fig. S1a). Fig. S1b shows the corresponding time-resolved (delayed) photoluminescence spectra, and the phosphorescence emission peaks of the SDP solution and solid powder

appeared at 450 and 533 nm, respectively. It indicates that phosphorescence emission can blue-shift in a single-molecule state compared to its aggregated state, agreeing well with the previous reports.^[5] The main peaks in the delayed spectra are assigned to the $n\text{-}\pi^*$ transition-dominated phosphorescence from the triplet excited state (T_1) of the compound. The intramolecular hydrogen bonding and $\pi\text{-}\pi$ stacking in the aggregated state (powder) induce the stabilization of the $n\text{-}\pi^*$ triplet state, leading to a red shift of the main peak compared with the monomeric state (dilute solution).



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Fig. 1. (a) Chemical structures of SDP and EVOH. (b, c) Photographs of the long-lived phosphorescence emission of SDP-EV^m and SDP-EV^s before and after turning off the UV-365nm excitation (C_d : 1%). Scale bar: 2cm. (d) Delayed phosphorescence spectra of SDP-EV^m and SDP-EV^s with a delay time of 10ms. Excitation by 365nm. (e) Chemical structure, frontier molecular orbital (HOMO/LUMO) distributions, and Jablonski energy level diagram of the SDP phosphor.

Next, the photophysical properties of SDP-doped EVOH are investigated. Melt-fabricated SDP-doped EVOH was prepared by using a single screw extruder at 180°C (Fig. S2 and Movie S1), and the extruded filaments were collected for further investigation (termed as SDP-EV^m). As a comparison, thin films of SDP-doped EVOH were also prepared by casting solution onto glass slides (termed as SDP-EV^s). The doping concentration (C_d) is 1%. Fig. 1b shows the photographs of SDP-EV^m filament before and after turning off UV-365nm excitation under the ambient conditions, in which an obviously blue afterglow with duration time of ~0.4s can be observed. In contrast, the film of SDP-EV^s exhibited green afterglow after the 365nm UV excitation (Fig. 1c). The delayed spectrum of SDP-EV^m filament exhibits a main emission at 455nm and a shoulder at 396nm. In contrast, an emission band at 510nm can be observed in SDP-EV^s film (Fig. 1d). These results reveal that the

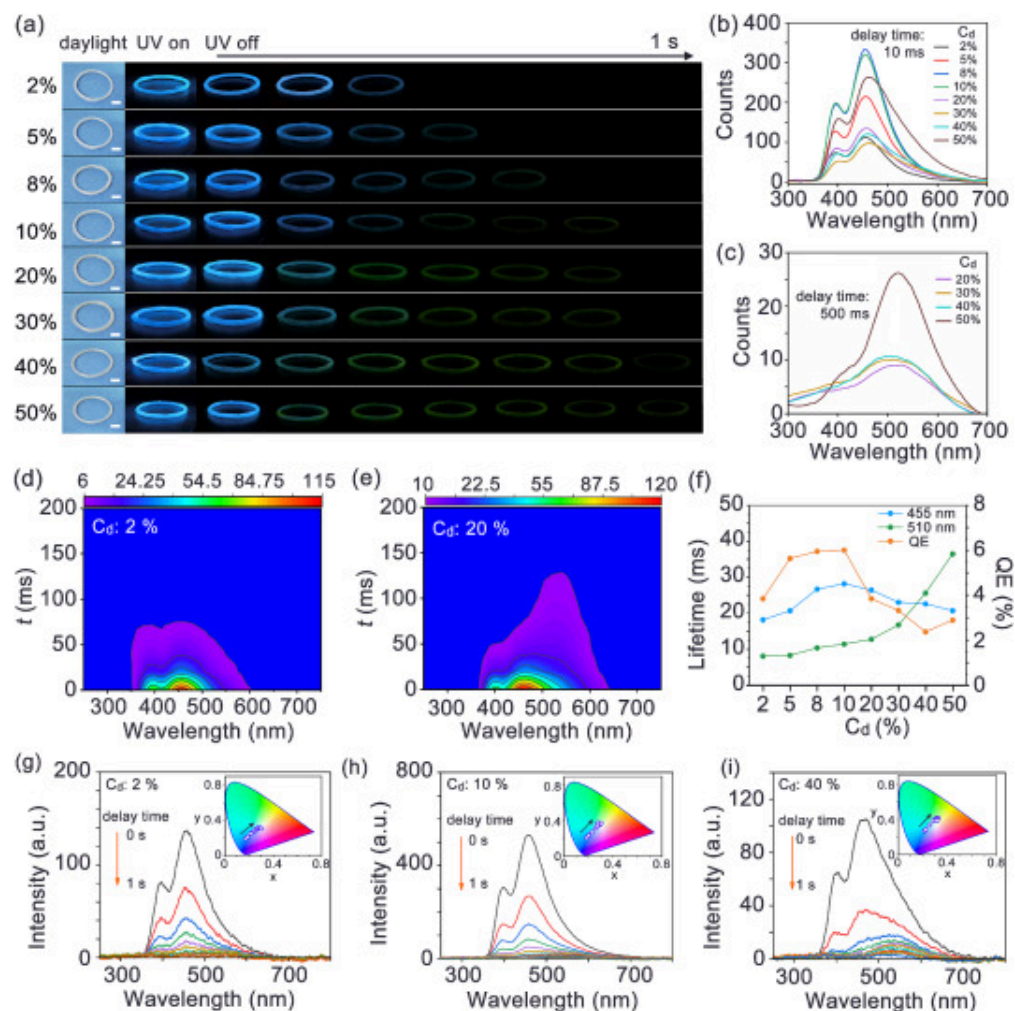
phosphorescence emissions of SDP-doped EVOH are determined by the processing method. The blue phosphorescence of SDP-EV^m is due to the single-molecule triplet emission of phosphor, whereas the green afterglow of SDP-EV^s is ascribed to the aggregation of phosphor molecules. For SDP-EV^m, the main emission peak was assigned to T₁-state phosphorescence dominated by n- π^* transitions, and the shoulder peak corresponded to T₂-state phosphorescence from π - π^* transitions, revealing the coexistence of both dispersed and aggregated phosphors. Compared with SDP-EV^m, intermolecular hydrogen bonding and π - π stacking in the aggregated state, causing a red shift of the main peak in SDP-EV^s.

Moreover, the photophysical properties of SDP-EV^m and SDP-EV^s were investigated. Fig. S3 shows the delayed emission spectra of SDP-EV^m and SDP-EV^s acquired under 254nm and 310nm excitation. Under 254nm excitation, only SDP-EV^m showed a distinct afterglow peak at ~427nm, while SDP-EV^s exhibited no measurable emission. Under 310nm excitation, both samples displayed strong afterglow with a main peak at ~493nm. This is attributed to aggregation-induced red-shifting of the phosphor's absorption: the more aggregated SDP-EV^m absorbs 254nm efficiently, whereas the dispersed SDP-EV^s does not. The afterglow peak blue-shifts from ~493nm to ~427nm with shorter excitation wavelength, consistent with higher-energy excitation yielding blue-shifted emission.

To reveal the photophysical origin of the RTP from SDP, we performed theoretical calculations to characterize its electronic structure and energy level characteristics (Fig. 1e). The sulfone (O=S=O) group in the SDP molecule acts as a strong electron-withdrawing unit to enhance spin-orbit coupling (SOC), which is critical for promoting intersystem crossing (ISC) and triplet exciton generation. Frontier molecular orbital analysis shows that the HOMO of SDP is mainly distributed on the benzene rings and hydroxyl groups, while the LUMO is localized on the central sulfone moiety. This separated orbital distribution reduces the singlet-triplet energy gap (ΔE_{ST}). As shown in the Jablonski diagram, the energy gap between the ground state (S_0) and the first singlet excited state (S_1) of SDP is 4.79eV, and the ΔE_{ST} between S_1 and the lowest triplet excited state (T_1) is 1.29eV. This moderate ΔE_{ST} enables efficient ISC from singlet to triplet states, and the rigid hydrogen-bonding network provided by the polymer matrix further suppresses non-radiative deactivation of triplet excitons, ultimately realizing stable RTP emission of SDP.

To further explore the photophysical properties of SDP-EV^m, a series of samples with varied doping concentrations were prepared. The mixing temperature is 165°C. After the 365-nm excitation, afterglow emission can be observed in all the filaments of SDP-EV^m, independent on the doping concentration (Fig. 2a). With the increasing of doping concentration, the afterglow duration time increased from 0.4s to 1s. The emission intensity of blue RTP reached its maximum when the SDP concentration was

approximately 8%. In contrast, green afterglow was observed in the cast films of SDP-EV^s, even at a low doping concentration, e.g., 0.5% (Fig. S4). Moreover, interestingly, as the doping concentration is above 20%, the color of phosphorescence emission of SDP-EV^m changed from blue to green after about 0.5 s. This observation was further confirmed by the delayed spectra captured at delay time. The phosphorescence spectra of SDP-EV^m collected at delay time of 10 ms exhibited an emission peak at 455 nm (Fig. 2b), whereas the spectra collected at delay time of 500 ms shown an emission peak at 530 nm (Fig. 2c). Such change of phosphorescence color was further confirmed by the time-resolved emission spectra of SDP-EV^m excited by 365 nm at room temperature (Fig. 2d and e and Fig. S5).



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Fig. 2. (a) Photographs of SDP-EV^m with different values of C_d before and after turning off the UV-365 nm excitation (Scale bar: 2 cm). (b, c) The delayed emission spectra of SDP-EV^m with delay time of 10 ms and 500 ms. Excitation by 365 nm. (d, e) Time-resolved afterglow mapping of SDP-EV^m with C_d of 2% and 20%. Excitation by 365 nm. (f) The phosphorescence emission lifetime and QE. (g-i) Time-resolved emission spectra with C_d of 2%, 10%, and 40% (inset: The CIE chromaticity diagrams). Excitation by 365 nm.

Furthermore, UV–vis absorption spectra of SDP-EV^m with varied C_ds were recorded (Fig. S6). The absorption band intensity at about 300 and 380nm gradually increases upon increasing the doping concentration, which may be ascribed to the π - π^* transition. The red-shifting of absorption peaks was observed with the increasing SDP concentration.

Meanwhile, the fluorescence emission spectra of SDP-EV^m displayed a wide emission peak at around 395nm (Fig. S7). The main emission of doped polymers is the same to pure SDP, but having a shoulder at 460nm. These results confirm that the dual emission of SDP-EV^m results from the dispersion of phosphors, rather than the formation of new chromophore during high-temperature mixing. Such multicolor afterglow regulated by isolated-molecular and aggregated-state phosphorescence is consistent with the previous report.^[32] It has been reported that phosphorescence from both isolated molecules and aggregated states can be achieved through oxygen consumption and the formation of a more rigid hydrogen-bond network. In the aggregated state, chromophores are shielded by surrounding molecules via tight molecular packing, which effectively suppresses oxygen-induced quenching and stabilizes triplet excitons. In contrast, isolated molecules are more prone to oxygen exposure, leading to more pronounced differences in phosphorescence emission before and after photoactivation compared to aggregates. Moreover, the lifetime of short-wavelength phosphorescence first increased and then decreased with the doping concentration (around 20–30ms), whereas the lifetime of long-wavelength phosphorescence increased with the doping concentration, i.e., from 9ms to 38ms (Fig. 2f and Fig. S8). In addition, the quantum efficiency (QE) of SDP-EV^m is shown in Fig. 2f. The decrease of QE with C_d is ascribed to the aggregation caused quenching (ACQ).

Figs. 2g–2i show the Commission International de l'Eclairage (CIE) coordinate diagrams calculated from the corresponding delayed phosphorescence emission spectra of SDP-EV^m with different doping concentration. The evolution of RTP colors with delay time was labeled in the CIE coordinates. For SDP-EV^m with doping concentration of 2%, the coordinate moved from (0.17, 0.19) to (0.29, 0.30) within 1 s (Fig. 2g). For SDP-EV^m with a

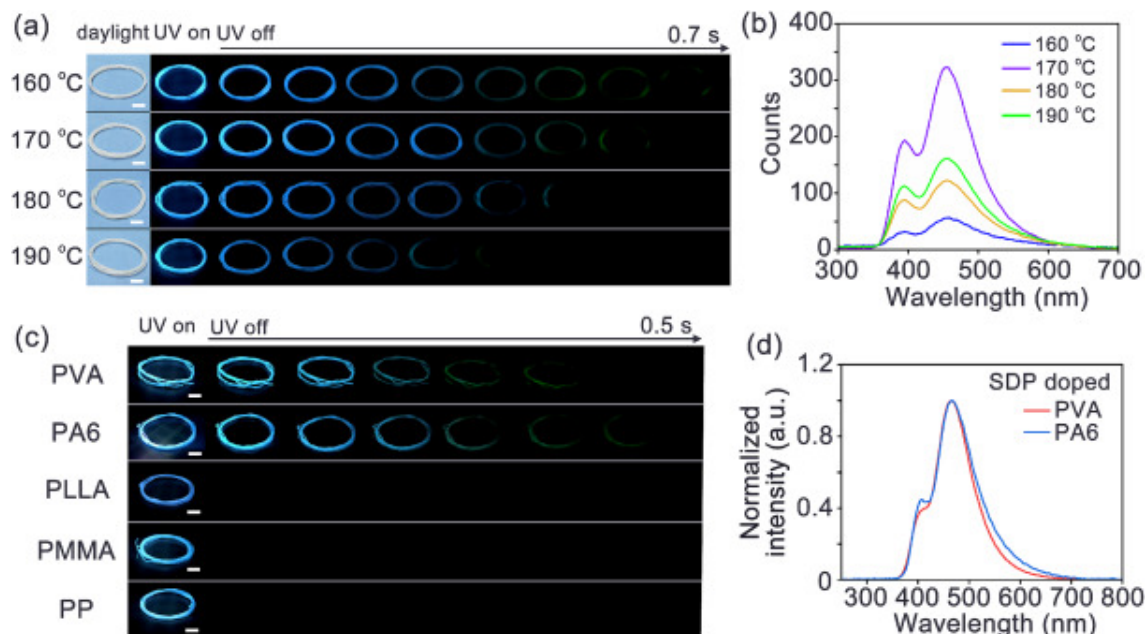
higher doping concentration, like 40%, the coordinate moved from (0.21, 0.27) to (0.32, 0.41) within 1 s, indicating an obvious color changing from blue to green (Fig. 2i).

The above results reveal that the time-dependent phosphorescence emission of SDP-EV^m is attributed to the dispersion of phosphors in polymer matrix. Due to the high temperature and strong shearing force, homogeneous dispersion of SDP molecules can be obtained during the process of melt mixing. The dispersed single molecule of SDP gives rises to the blue phosphorescence, similar to the case in the dilute solution of SDP. In contract, the aggregation of SDP molecules during solvent evaporation leads to the green afterglow of SDP-EV^s films. However, as the doping concentration is higher than the saturation concentration, aggregation of SDP takes place even in the melt-processed composites. Therefore, the mixed emission with both blue and green can be observed in the SDP-EV^m with high doping concentration.

Besides the processing method, interactions between phosphor and polymeric matrix are also critical to the dispersion of phosphors. To investigate the interactions between SDP and EVOH matrix, SDP-EV^m is further explored by FTIR. The presence of broad band at 3320 cm⁻¹ assigning to the stretching vibration of OH groups indicate the formation of extensive inter-molecular/intra-molecular hydrogen bonds, which can improve the 'dissolving' of SDP into the polymer matrix (Fig. S9). In addition, with increasing of doping concentration, the intensity of stretching vibration of sulfone group (O=S=O) at 1146 cm⁻¹ gradually increased, and the intensity of hydrogen bond decreased. This result is consistent with the previous observation, namely, at high doping concentration, aggregation of SDP molecules takes place, resulting the green phosphorescence.^[33]

To further understand the influence of processing condition on the dispersion of SDP in EVOH, RTP emissions of SDP-EV^m (C_d: 0.5%) prepared at different processing temperatures was investigated. Beside blue RTP emissions, green afterglow gradually faded away with the processing temperature (Fig. 3a). In particular, only blue afterglow can be observed in the SDP-EV^m prepared at 180 °C. However, as the mixing temperature exceeded 180 °C, the reduction of afterglow duration can be observed. Note that the

melting point (T_m) of EVOH used in the present work is 181 °C (Fig. S10a). Fig. 3b shows the delayed emission spectra of SDP-EV^m prepared at different processing temperatures (delay time: 10ms). Notably, the emission intensity exhibits a non-monotonic variation with temperature, reaching the maximum at 170 °C and gradually decreasing as the temperature rises to 190 °C. For one thing, the above results reveal that the increased mixing temperature can enhance the dispersion of phosphors, suppressing the long-wavelength RTP emission. For the other thing, as the processing.



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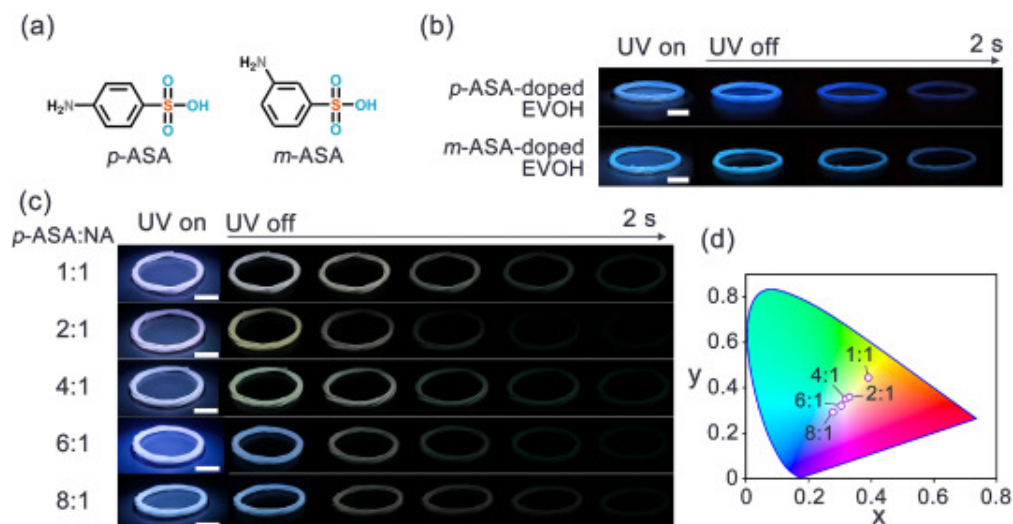
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Fig. 3. (a) Photographs of SDP-EV^m prepared at different processing temperatures before and after turning off the UV-365 nm excitation (C_d :0.5%). Scale bar: 2 cm. (b) The delayed emission spectra of SDP-EV^m with delay time of 10ms. Excitation by 365 nm. (c) Photographs of SDP-doped PVA, PA6, PLLA, PMMA and PP before and after turning off the UV-365 nm excitation. Scale bar: 2 cm. (d) The delayed emission spectra of SDP-doped PVA and PA6. Excitation by 365 nm.

temperature exceeds 180°C, degradation of EVOH will take place, resulting in the weakening of RTP. As shown in Figs. S10b and S10c, weight loss of pure EVOH and SDP-EV^m can be observed above 180°C in the TGA curves. The above results also reveal that SDP-EV^m with pure blue phosphorescence emission could be achieved under an appropriate condition.

The above results uncover that blue RTP emission can be achieved in SDP-doped EVOH via melt processing. To verify whether the blue phosphorescence of SDP can be obtained in other matrices, we prepared SDP-doped polyvinyl alcohol (PVA), polyamide 6 (PA6), poly(L-lactide) (PLLA), poly(methyl methacrylate) (PMMA), and polypropylene (PP) by using a single-screw extruder (C_d : 5%). It is found that blue afterglow can be observed in the SDP-doped PVA and PA6 after the UV-365nm light was turned off under ambient conditions, however, no afterglow can be observed in SDP-doped PMMA, PLLA, and PP (Fig. 3c). The delayed photoluminescence spectra of SDP-doped PVA and PA6 exhibit main peak at 465nm and should peak at 408nm, which is similar to that of SDP-EV^m (Fig. 3d). It suggests that hydrogen bonding between SDP and the polymeric matrices is crucial to the generation of blue RTP emissions, which can stabilize the triplet exciton.

To demonstrate that the combination of EVOH matrix and melt-mixing provide an effective and widely applicable method to prepare for various phosphors, EVOH doped with *p*-ASA and *m*-ASA were prepared via melt mixing. The doping concentration is 2%, and the processing temperature is 170°C (Fig. 4a). With the excitation by 254-nm UV light, blue afterglow with duration time of ~2s can be observed from the filaments of *p*-ASA-doped EVOH and *m*-ASA-doped EVOH under the ambient condition (Fig. 4b). According to their delayed phosphorescence spectra, *p*-ASA-doped EVOH and *m*-ASA-doped EVOH present dominant emission bands at 433nm and 459nm at 293K, respectively (Fig. S11). In contrast, solution-casted film of *p*-ASA-doped EVOH exhibited weak green afterglow, and no visible afterglow can be seen in the casted film of *m*-ASA-doped EVOH, maybe due to the aggregation induced quenching of triplet exciton (Fig. S12). These results confirm the universality of our approach for polymeric materials with blue RTP emission.



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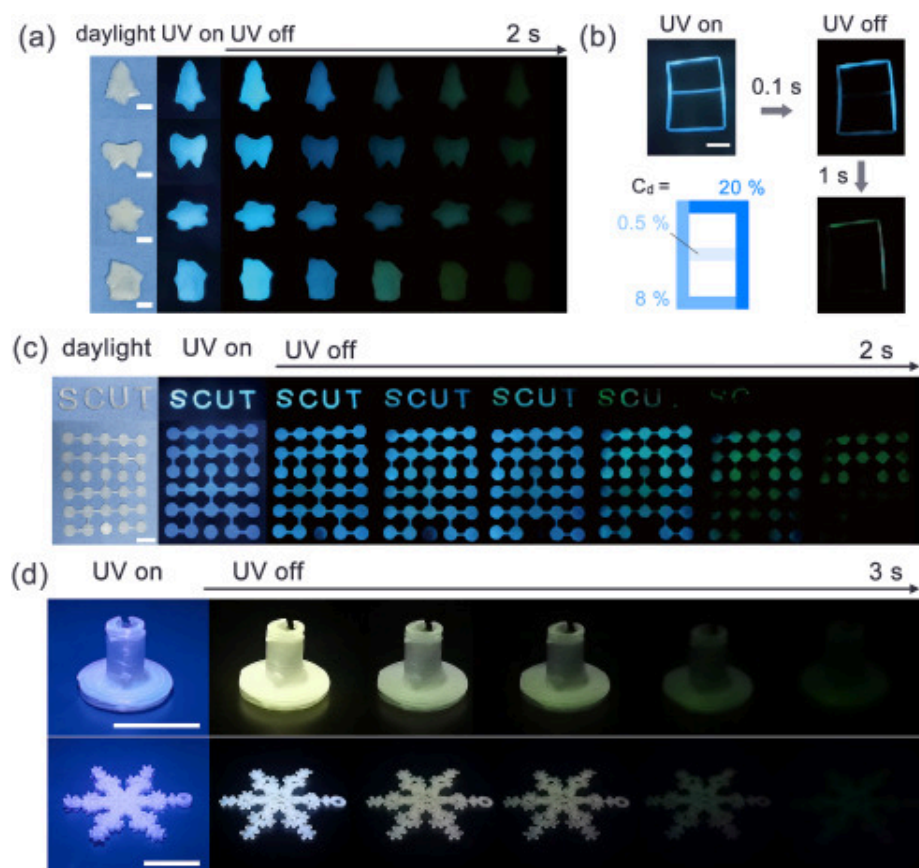
Fig. 4. (a) Chemical structures of *p*-ASA and *m*-ASA. (b) Photographs of *p*-ASA-doped EVOH and *m*-ASA-doped EVOH before and after turning off the UV-254nm excitation. Scale bar: 2cm. (c) Photographs of EVOH doped with *p*-ASA and NA before and after turning off the UV-254nm excitation. Scale bar: 2cm. (d) CIE chromaticity diagram of EVOH doped with *p*-ASA and NA at different ratios.

We already have developed the method for the preparation of blue phosphorescence. Moreover, it also provides an opportunity to achieve white RTP emission by superposing blue and yellow phosphorescence.^[34] To this end, NA was selected as a counterpart with complementary phosphorescence color. NA-doped EVOH prepared via melting process exhibits yellow afterglow with duration time of ~2s was visible in the NA-doped EVOH after the UV-254nm light was turned off under ambient condition (Figs. S13 and S14). Then, EVOH doped with *p*-ASA and NA was prepared by one-step mixing with an extruder. Impressively, whiteish afterglow with duration time of ~2s can be observed in the *p*-ASA/NA-doped EVOH after the 254nm UV excitation (Fig. 4c). The CIE chromaticity diagram of *p*-ASA/NA-doped EVOH indicates that the color of phosphorescence emission

is determined by the ratio of *p*-ASA and NA (Fig. 4d). Namely, the phosphorescence color changed from blue (0.28, 0.29) to yellow (0.39, 0.44) with good linearity of the CIE coordinates as the ratio of *p*-ASA/NA increased from 1:1 to 8:1. In particular, as the ratio of *p*-ASA/NA reaches 2:1, the CIE coordinates of doped EVOH (0.33, 0.36) are nearly equal to the standard equal energy white light emission (0.33, 0.33) (Movie S2). To evaluate energy transfer between *p*-ASA and NA during white-light emission construction, we analyzed spectral overlap and phosphorescence lifetime changes. The UV–Vis and fluorescence spectra show no overlap between *p*-ASA emission and NA absorption, violating the FRET condition. Phosphorescence decay lifetimes at 430nm (*p*-ASA) and 585nm (NA) remain unchanged across all mixing ratios, indicating no excited-state quenching or energy transfer (Fig. S15). White-light emission arises from the linear superposition of independent RTP emissions from *p*-ASA and NA, confirming no effective energy transfer occurs between the two phosphors. It also implies that both *p*-ASA and NA were well-dispersed in the matrix, therefore the isolation of chromophores effectively suppressed the exciton energy transfer process. It suggests that, not only white RTP, but other phosphorescence colors could also be prepared by multi-component doping via melt mixing.

The above results demonstrate the advantages of melt processing in controlling the emissions colors of RTP. Compared with the solution-mixing method, melt processing endows the RTP material with excellent processability and formability. In particular, EVOH is a typical thermoplastic polymer and has been widely used due to its outstanding processability. Thus, phosphor-doped EVOH can be further shaped via thermoplastic molding techniques for specific application scenarios. As shown in Fig. 5a, SDP-EV^m can be molded into various three-dimensional (3D) shapes, and under 365nm UV excitation, phosphorescence emissions can be observed. Moreover, by merely tuning the doping concentration, more complex information can be obtained from the same polymer. For example, several SDP-EV^m bars with different SDP doping concentrations were arranged into a shape of digit '8' (Fig. 5b). Under 365nm UV excitation, the encryption model displays the letter '8' in blue. After the UV light is turned off, the afterglow of the model

sequentially transitions from the shape of '8' to '0' and then to '7', verifying its feasibility for anti-counterfeiting applications.



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Fig. 5. Demonstration of applications based on blue and white RTP materials. (a) Afterglow emission of injection-molded RTP material samples (tree, tooth, star, and irregular shapes) under 365nm UV excitation, with sequential photographs recorded from UV on to 2s after UV off. (b) Luminescence evolution of 3D-printed RTP square patterns (with dye doping concentrations of 0.5%, 8%, and 20%) under UV on and at 0.1 s and 1 s after UV off. (c) Digital information anti-counterfeiting and encryption realized by adjusting dye doping ratios, showing the "SCUT" text and patterned arrays under daylight,

UV on, and 2s after UV off. (d) Long-lived afterglow of 3D-printed complex objects (candle-shaped and snowflake-shaped samples) under UV on and at sequential time points up to 3s after UV off. Scale bar: 2cm. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

In addition, extruded phosphor-doped EVOH filaments can be directly used for 3D printing. As demonstrated in Fig. 5c, a 3D-printed pattern including a 'SCUT' and a pattern exhibit bright blue fluorescence under 365-nm UV excitation. After the UV light is turned off, a phosphorescent afterglow that transitions from blue to green is clearly observable. Moreover, Fig. 5d shows a 3D-printed candle holder and a snowflake with *p*-ASA/NA-doped EVOH, and after 254-nm UV excitation, white afterglow with duration time of ~2s can be observed (Movies S3 and S4).

4. Conclusions

This work proposes a scalable melt-processing strategy to address these issues by dispersing phosphors (e.g., SDP, *p*-ASA, *m*-ASA) in hydrogen-bonding polymer matrices (e.g., EVOH, PVA, PA6). Melt processing enables uniform molecular dispersion via high temperature and shear forces, while matrix-phosphor hydrogen bonding stabilizes triplet states, achieving single-molecule-derived blue RTP at 455 nm, which is distinct from the green emission of aggregated counterparts. The strategy offers exceptional tunability, namely, adjusting doping concentration or phosphor ratios enables dynamic color transitions (blue-to-green) and white RTP (CIE: 0.33, 0.36). Moreover, the materials exhibit excellent processability for injection molding and 3D printing, with demonstrated applications in anti-counterfeiting via pattern transitions and optoelectronic patterning. This work provides a facile, scalable route to controlled-luminescence blue RTP materials, advancing fundamental RTP mechanisms and expanding their practical use in encryption, displays, and beyond.

CRedit authorship contribution statement

Jintong Wu: Investigation. **Chengyue Wu:** Investigation, Writing – original draft. **Jiahao Liang:** Investigation, Writing – original draft. **Jiaer Zhu:** Investigation. **Junhong Liang:** Investigation, Writing – original draft. **Yingying Wang:** Funding acquisition, Writing – review & editing. **Tao Wen:** Writing – review & editing, Funding acquisition, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

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Supplementary material



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Supplementary Video 3



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Supplementary Video 4

Data availability

Data will be made available on request.

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